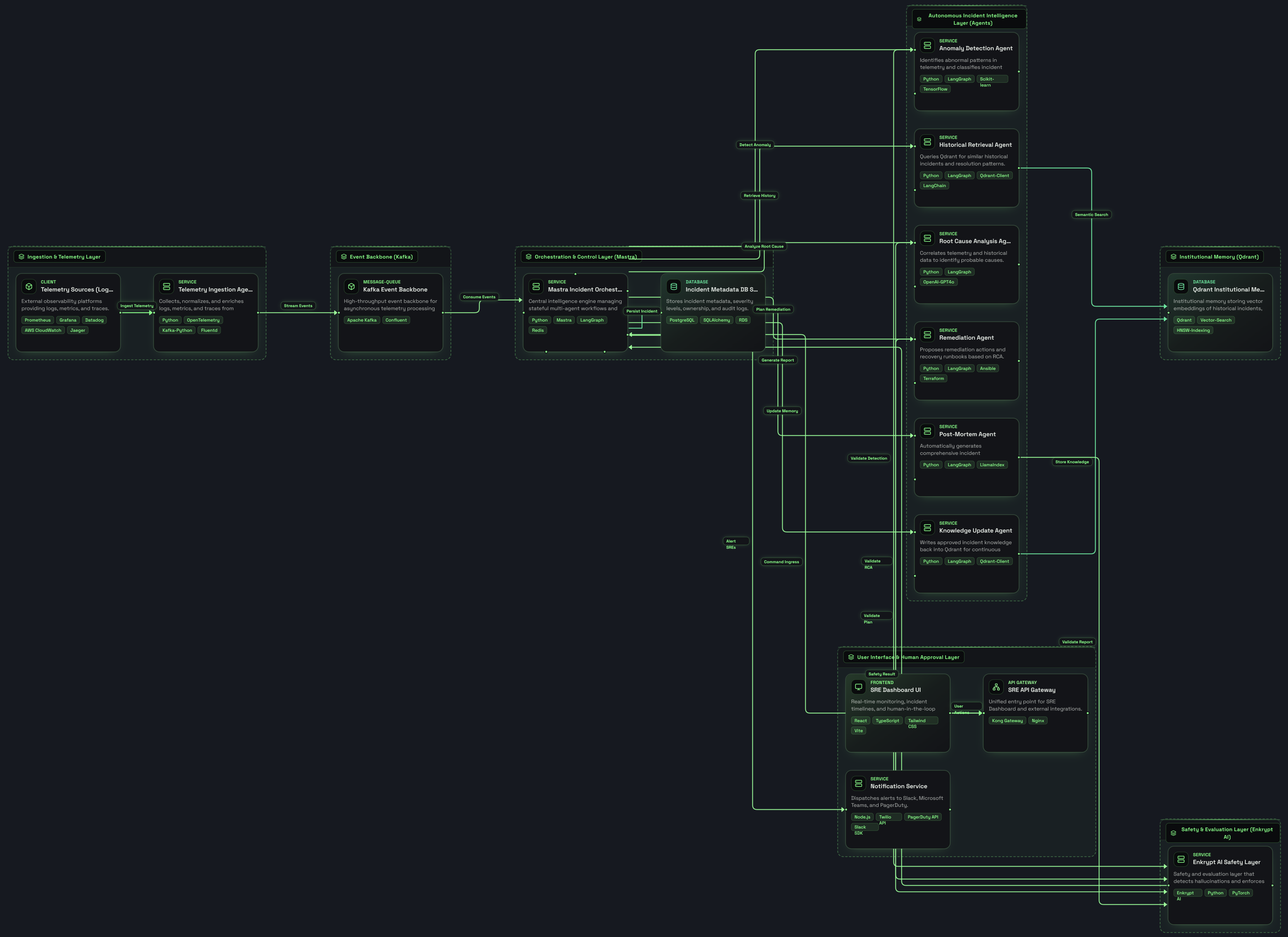
Ingestion & Telemetry Layer
CLIENT
Telemetry Sources (Log...
External observability platforms providing logs, metrics, and traces.
Prometheus
Grafana
Datadog
AWS CloudWatch
Jaeger
Ingest Telemetry
SERVICE
Telemetry Ingestion Age...
Collects, normalizes, and enriches logs, metrics, and traces from
Python
OpenTelemetry
Kafka-Python
Fluentd
Stream Events
Event Backbone (Kafka)
MESSAGE-QUEUE
Kafka Event Backbone
High-throughput event backbone for asynchronous telemetry processing
Apache Kafka
Confluent
Consume Events
Orchestration & Control Layer (Mastra)
SERVICE
Mastra Incident Orchest...
Central intelligence engine managing stateful multi-agent workflows and
Python
Mastra
LangGraph
Redis
Persist Incident
DATABASE
Incident Metadata DB S...
Stores incident metadata, severity levels, ownership, and audit logs.
PostgreSQL
SQLAlchemy
RDS
Plan Remediation
Generate Report
Update Memory
Validate Detection
Alert SREs
Command Ingress
Validate RCA
Validate Plan
Autonomous Incident Intelligence Layer (Agents)
SERVICE
Anomaly Detection Agent
Identifies abnormal patterns in telemetry and classifies incident
Python
LangGraph
Scikit-learn
TensorFlow
Detect Anomaly
Retrieve History
SERVICE
Historical Retrieval Agent
Queries Qdrant for similar historical incidents and resolution patterns.
Python
LangGraph
Qdrant-Client
LangChain
Analyze Root Cause
SERVICE
Root Cause Analysis Ag...
Correlates telemetry and historical data to identify probable causes.
Python
LangGraph
OpenAI-GPT4o
SERVICE
Remediation Agent
Proposes remediation actions and recovery runbooks based on RCA.
Python
LangGraph
Ansible
Terraform
SERVICE
Post-Mortem Agent
Automatically generates comprehensive incident
Python
LangGraph
LlamaIndex
Store Knowledge
SERVICE
Knowledge Update Agent
Writes approved incident knowledge back into Qdrant for continuous
Python
LangGraph
Qdrant-Client
Semantic Search
Institutional Memory (Qdrant)
DATABASE
Qdrant Institutional Me...
Institutional memory storing vector embeddings of historical incidents,
Qdrant
Vector-Search
HNSW-Indexing
Validate Report
User Interface & Human Approval Layer
Safety Result
FRONTEND
SRE Dashboard UI
Real-time monitoring, incident timelines, and human-in-the-loop
React
TypeScript
Tailwind CSS
Vite
User Actions
API GATEWAY
SRE API Gateway
Unified entry point for SRE Dashboard and external integrations.
Kong Gateway
Nginx
SERVICE
Notification Service
Dispatches alerts to Slack, Microsoft Teams, and PagerDuty.
Node.js
Twilio API
PagerDuty API
Slack SDK
Safety & Evaluation Layer (Enkrypt AI)
SERVICE
Enkrypt AI Safety Layer
Safety and evaluation layer that detects hallucinations and enforces
Enkrypt AI
Python
PyTorch